
EXERCISE

Lecture 10: Routing cont.

AI508 — Algorithmic Game Theory

Simon Holm
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1 Recall

$$\alpha(G) = \sup_{c \in G} \sup_{r > 0} \max_{0 \leq x \leq r} \frac{r \cdot c(r)}{x \cdot c(x) + (r - x) \cdot c(x)}$$

$\leq r$ is actually irrelevant since the fraction becomes tiny when $x \gg r$ me

where c : is continuous and nonnegative

Key Theorem

For every set G of cost functions & for every selfish routing network with

$$c_i \in G \quad \forall i$$

Then

$$\text{PoA} \leq \alpha(G) \text{ despite } \alpha \text{ only defined on Pigov networks}$$

2 Paths and traffic

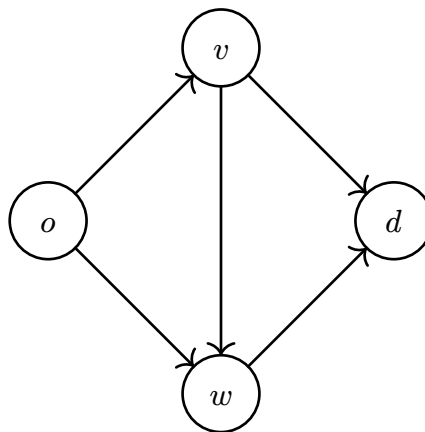


Figure 1: A modified traffic system

Directed : (V, E)

And

$$o \rightarrow v : c(x) = x$$

$$o \rightarrow w : c(x) = 1$$

$$v \rightarrow d : c(x) = 1$$

$$w \rightarrow d : c(x) = x$$

$$v \rightarrow w : c(x) = 0$$

With r unit for total demand going from $o \rightarrow d$

For every given path p , f_p denotes the fraction of traffic using path p , where

$$\sum_{p \in \mathcal{P}} f_p = r$$

Given an edge e then,

$$f_e = \sum_{p \in \mathcal{P}: e \in p} f_p$$

Example: computing f_e

Given g :

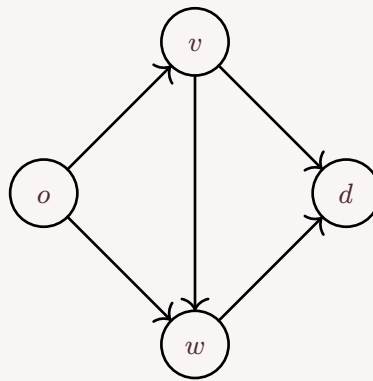


Figure 2: A modified traffic system

$$g : (V, E)$$

Where

$$o \rightarrow v \rightarrow d : 1/4$$

$$o \rightarrow w \rightarrow d : 1/4$$

$$o \rightarrow v \rightarrow w \rightarrow d : 1/2$$

Then

$$f_{(o,v)} = \sum_{p \in \mathcal{P}: e \in p} f_p = \frac{3}{4}$$

3 Equilibrium flow

Definition: Equilibrium flow

A flow f is said to be an equilibrium if

$$f_{\hat{p}}^{\text{eq}} > 0 \implies \hat{p} \in \arg \min_{p \in \mathcal{P}} \left\{ \sum_{e \in p} c_e(f_e) \right\}$$

Example: Optimal cost

Given a Braess network g

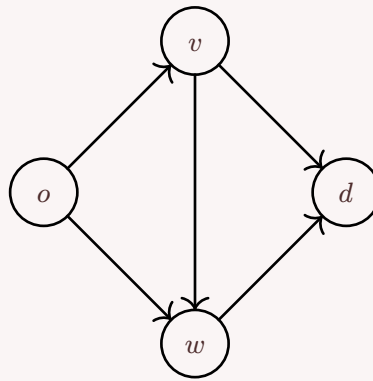


Figure 3: A modified traffic system

$$g : (V, E)$$

Where

$$o \rightarrow v : c(x) = x$$

$$o \rightarrow w : c(x) = 1$$

$$v \rightarrow d : c(x) = 1$$

$$w \rightarrow d : c(x) = x$$

$$v \rightarrow w : c(x) = 0$$

Is the flow shown above an equilibrium flow?

Then

$$c_{o,v,d}(f) = c_{o,v}\left(\frac{3}{4}\right) + c_{w,d}\left(\frac{3}{4}\right) = \frac{3}{4} + \frac{3}{4} = \frac{3}{2}$$

$$c_{o,w,d}(f) = c_{o,w}\left(\frac{1}{4}\right) + c_{w,d}\left(\frac{3}{4}\right) = 1 + \frac{3}{4}$$

Since $3/2 \neq 1 + 3/4$, the flow is **not** an equilibrium.

4 Optimal Flow

For any flow f

$$\text{Cost}(f) = \underbrace{\sum_{p \in \mathcal{P}} f_p \cdot c_p(f)}_{\text{via paths}} = \underbrace{\sum_{e \in E} f_e \cdot c_e(f_e)}_{\text{via edges}}$$

The optimal flow f^* is then defined as

$$f^* \in \arg \min_{f \text{ feasible}} \text{Cost}(f)$$

and

$$\text{PoA} = \frac{\text{Cost}(f^{\text{eq}})}{\text{Cost}(f^*)} \leq \alpha(G)$$

Take the equilibrium flow f^{eq} and let g be any feasible flow. The following inequality holds.

$$\sum_{e \in E} (g_e - f_e^{\text{eq}}) \cdot c_e(f_e^{\text{eq}}) \geq 0$$

Then by the definition of $\alpha(G)$,

$$\alpha(G) \geq \frac{r \cdot c(r)}{x \cdot c(x) + (r - x)c(r)}$$

and therefore

$$x \cdot c(x) \geq \frac{1}{\alpha(G)} \cdot r \cdot c(r) + (x - r)c(r)$$

Then let

$$c = c_e, \quad r = f_e^{\text{eq}}, \quad x = f_e^* \quad \forall e \in E$$

then

$$\begin{aligned} f_e^* c_e(f_e^*) &\geq \frac{1}{\alpha(G)} f_e^{\text{eq}} c_e(f_e^{\text{eq}}) + (f_e^* - f_e^{\text{eq}}) c_e(f_e^{\text{eq}}) \\ \text{Cost}(f^*) &\geq \frac{1}{\alpha(G)} \text{Cost}(f^{\text{eq}}) + \underbrace{\sum_{e \in E} (f_e^* - f_e^{\text{eq}}) \cdot c_e(f_e^{\text{eq}})}_{\geq 0} \end{aligned}$$

Since the sum on *r.h.s.* is nonzero, it cannot be a contributing factor to the fact that

$$\text{Cost}(f^*) \geq \frac{1}{\alpha(G)} \text{Cost}(f^{\text{eq}})$$

5 Over provisioned

Given a cost function c

$$c(x) = \begin{cases} \frac{1}{u_e - x} & \text{if } x < u_e \\ +\infty & \text{otherwise} \end{cases}$$

where $u_e > 0$ is the total capacity of an edgy

Definition: Over provisioned

Fix $\beta \in (0, 1)$ A routing network is β -over provisioned if at the equilibrium

$$f_e^{\text{eq}} \leq (1 - \beta) \cdot v_e \quad \forall e \in E$$

Also saying. A network is overprovisioned if too much capacity is *unused*.

Example: Pigov example

Let g be a Pigov network

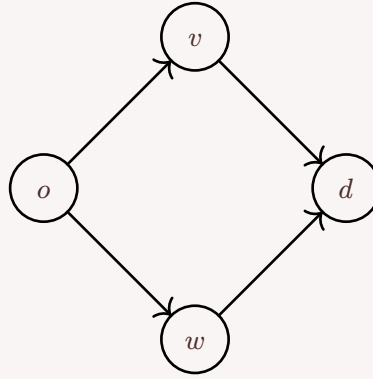


Figure 4: A Pigov network for example

Where

$$o \rightarrow v \rightarrow d : c(x) = \frac{1}{u - r}$$

$$o \rightarrow w \rightarrow d : c(x) = \frac{1}{u - x}$$

Lets set $r = (1 - \beta) \cdot u$, and bound PoA as following

$$\text{PoA} \leq \frac{1}{2} \left(1 + \frac{1}{\sqrt{\beta}} \right)$$

As $\beta \rightarrow 1$ then $\text{PoA} \rightarrow 1$

As $\beta \rightarrow 0$ then $\text{PoA} \rightarrow \infty$